Jiaxin Guan

Applied Cryptographer, Zellic

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Education

- **Princeton University** **Princeton, NJ**
  PhD and M.A. in Computer Science *09/17 – 07/23*
  - Research Area: Cryptography
  - Thesis: *Cryptography against Space-Bounded Adversaries*
- **Stanford University** **Stanford, CA**
  M.S. and B.S. with Honors in Computer Science - GPA: 3.9/4.3 *09/13 – 06/17*

Academic Interests

AI Security & Safety, Model Alignment, Agentic Identity, LLM Watermarking, Post-Quantum Cryptography, Information-Theoretic Cryptography, Space-Bounded Cryptography, Functional Encryption, Coding Theory, Time-Space Lower Bounds, Complexity Theory, Pseudorandomness, Meta-Complexity

Publications

1. Jiaxin Guan and Daniel Wichs, "**Streaming One-Time Programs in the Bounded Storage Model**", In Preparation
2. Marshall Ball and Jiaxin Guan, "**A Complexity Theoretic Approach to Proofs of Space**", TCC 2026 ***(Outstanding Paper Award)***
3. Pratish Datta, Jiaxin Guan, Alexis Korb, and Amit Sahai, "**Multi-Hop Functional Encryption**", TCC 2026
4. Pratish Datta, Jiaxin Guan, Alexis Korb, and Amit Sahai, "**(Multi-Input) FE for Randomized Functionalities, Revisited**", TCC 2025
5. Pratish Datta, Jiaxin Guan, Alexis Korb, and Amit Sahai, "**Adaptively Secure Streaming Functional Encryption**", TCC 2025
6. Yevgeniy Dodis, Jiaxin Guan, Peter Hall, and Alison Lin, "**HELP: Everlasting Privacy through Server-Aided Randomness**", IACR CiC Volume 1 (2024), Issue 4
7. Jiaxin Guan and Hart Montgomery, "**On Sequential Functions and Fine-Grained Cryptography**", CRYPTO 2024
8. Jiaxin Guan, Daniel Wichs, and Mark Zhandry, "**Multi-Instance Randomness Extraction and Security against Bounded-Storage Mass Surveillance**", TCC 2023
9. Jiaxin Guan, Alexis Korb, and Amit Sahai, "**Streaming Functional Encryption**", CRYPTO 2023
10. Dan Boneh, Jiaxin Guan, and Mark Zhandry, "**A Lower Bound on the Length of Signatures based on Group Actions and Generic Isogenies**", EUROCRYPT 2023

11. Jiaxin Guan, Daniel Wichs, and Mark Zhandry, "**Incompressible Cryptography**", EUROCRYPT 2022
12. Jiaxin Guan and Mark Zhandry, "**Iterated Inhomogeneous Polynomials**", CFail 2021
13. Jiaxin Guan and Mark Zhandry, "**Disappearing Cryptography in the Bounded Storage Model**", TCC 2021
14. Jiaxin Guan and Mark Zhandry, "**Simple Schemes in the Bounded Storage Model**", EUROCRYPT 2019
15. James Bartusek, Jiaxin Guan, Fermi Ma, and Mark Zhandry, "**Return of GGH15: Provable Security Against Zeroizing Attacks**", TCC 2018

Talks

1. (Multi-Input) FE for Randomized Functionalities, Revisited
   - TCC 2025 Conference Talk (December 2025)
2. Space Jammed: Cryptography against Space-Bounded Adversaries
   - National University of Singapore (March 2025)
   - New Jersey Institute of Technology (March 2025)
   - Queens College Q4C Colloquium (December 2024)
3. On Sequential Functions and Fine-Grained Cryptography
   - CRYPTO 2024 Conference Talk (August 2024)
4. Multi-Instance Randomness Extraction and Security against Bounded-Storage Mass Surveillance
   - ITC 2024 Highlights Track (August 2024)
   - SJTU John Hopcroft Center Lecture Series (January 2024)
   - NYU Crypto Reading Group (December 2023)
   - TCC 2023 Conference Talk (December 2023)
5. A Lower Bound on the Length of Signatures based on Group Actions and Generic Isogenies
   - EUROCRYPT 2023 Conference Talk (April 2023)
   - CMU CyLab Crypto Seminar (April 2023)
   - Texas Crypto Day (April 2023)
6. Incompressible Cryptography
   - NTT Research (July 2022)
   - EUROCRYPT 2022 Conference Talk (May 2022)
   - UCLA Crypto Reading Group (April 2022)
   - CMU CyLab Crypto Seminar (April 2022)
   - Stanford Security Seminar (March 2022)
7. Disappearing Cryptography and Incompressible Cryptography
   - NYU Crypto Reading Group (January 2022)
   - TCC 2021 In-Person Workshop Talk (November 2021)
8. Disappearing Cryptography in the Bounded Storage Model

- TCC 2021 Conference Talk (November 2021)

9. Iterated Inhomogeneous Polynomials
   - CFail 2021 Workshop, a CRYPTO 2021 Affiliated Event (August 2021)
10. Simple Schemes in the Bounded Storage Model
    - EUROCRYPT 2019 Conference Talk (May 2019)
    - Princeton General Exam (May 2019)

Professional Activities

Program Committee:

CRYPTO 25, **IACR CiC** 25

Conference Reviews:

CRYPTO 18, 26, **EUROCRYPT** 22, 23, 26, 27, **TCC** 21, 22, 23, 24, 25
ASIACRYPT 19, 20, 25, 26, **STOC** 22, **ITCS** 21, 25, **CCC** 24, **RANDOM** 25

Honors and Awards

- **Outstanding Paper Award**, TCC 2026, for "A Complexity Theoretic Approach to Proofs of Space" (with Marshall Ball)
- **Bug of the Month**, Zellic (Mar. 2026), for a selective failure attack against a production PIR implementation
- **Bug of the Month**, Zellic (Jan. 2026), for a critical MPC/BFT soundness vulnerability

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Work Experience

- **Zellic** **New York, NY**
  Applied Cryptographer *06/25 – Present*
  - Perform adversarial security audits of various cryptographic protocols and blockchain systems under realistic threat models, identifying exploitable flaws that arise from implementation bugs, specification gaps, and unsafe trust assumptions.
  - Conduct original research on cryptographic primitives and protocol security, applying formal methods and rigorous analysis to reason about system security, robustness, and potential attack vectors.
  - Received two company-wide "Bug of the Month" awards for vulnerabilities discovered during client audits: a critical MPC/BFT soundness vulnerability (Jan. 2026) and a selective failure attack against a production PIR implementation (Mar. 2026).
- **New York University** **New York, NY**
  Assistant Professor / Faculty Fellow *09/23 – 06/25*
  - Taught a high-enrollment section of *Basic Algorithms* as primary instructor, focusing on formal reasoning about correctness and computational complexity of algorithms.
  - Published original research at top cryptography venues (CRYPTO, TCC) on information-theoretic cryptography, functional encryption, and sequential functions.

Skills

Programming: Python, Rust, C++, C, Go
Tools: PyTorch, NumPy, Docker, React

Teaching Experience

New York University:

- Instructor: CSCI-UA.0310-005, Basic Algorithms, Spring 2025
- Instructor: CSCI-UA.0310-007, Basic Algorithms, Fall 2024
- Instructor: CSCI-UA.0310-005, Basic Algorithms, Spring 2024
- Instructor: CSCI-UA.0310-007, Basic Algorithms, Fall 2023

Princeton University:

- Assistant in Instruction: COS 533, Advanced Cryptography, Spring 2021
- Assistant in Instruction: COS 433, Cryptography, Spring 2020
- Assistant in Instruction: COS 445, Economics and Computation, Spring 2019
- Assistant in Instruction: COS 432, Information Security, Fall 2018

Stanford University:

- Teacher's Assistant: CS 155, Computer and Network Security, Spring 2017
- Senior Section Leader: CS 106B, Programming Abstractions, Winter 2016
- Senior Section Leader: CS 106X, Programming Abstractions (Accelerated), Fall 2015
- Senior Section Leader: CS 106B, Programming Abstractions, Spring 2015
- Senior Section Leader: CS 106X, Programming Abstractions (Accelerated), Winter 2015
- Senior Section Leader: CS 106X, Programming Abstractions (Accelerated), Fall 2014
- Section Leader: CS 106B, Programming Abstractions, Spring 2014
- Section Leader: CS 106A, Programming Methodology, Winter 2013

Patents

1. Pratish Datta, Jiaxin Guan, Alexis Korb, and Amit Sahai, "**Multi-Hop Functional Encryption with Selective Intermediate Disclosure and Forwardable Encrypted State**", US Provisional Patent Application 64/057,167, Pending (Filed May 2026)
2. Pratish Datta, Jiaxin Guan, Alexis Korb, and Amit Sahai, "**Adaptively Secure Randomized Multi-Input Functional Encryption System and Method**", US Patent Application 19/383,466, Pending (Filed Nov. 2025)
3. Yan Huang, Xinyi Han, and Jiaxin Guan, "**Cloud-Based Digital Image Provenance Analysis with Visual Difference Overlay Construction**", US Patent Application 19/290,154, Pending (Filed Aug. 2025)

4. Yan Huang, Xinyi Han, and Jiaxin Guan, "**Scalable Content Authentication and Licensing System**", US Provisional Patent Application 63/826,375, Pending (Filed Jun. 2025)
5. Pratish Datta, Jiaxin Guan, Alexis Korb, and Amit Sahai, "**Adaptively Secure Streaming Functional Encryption System and Method**", US Patent Application 19/064,398, Published (Filed Feb. 2025)
6. Yan Huang, Xinyi Han, Jiaxin Guan, and James Denaro, "**Distributed Content Authentication and Licensing System**", US Provisional Patent Application 63/754,034, Pending (Filed Feb. 2025)
7. Yan Huang, Xinyi Han, and Jiaxin Guan, "**Watermarking Digital Media for Content Verification with Differential Detection**", US Patent Application 18/966,712, Allowed (Filed Dec. 2024)
8. Yan Huang, Jiaxin Guan, and Xinyi Han, "**Watermarking Digital Media for Authenticated Content Verification**", PCT Patent Application PCT/US24/39021, Pending (Filed Jul. 2024)
9. Yan Huang, Xinyi Han, and Jiaxin Guan, "**Watermarking Digital Media for Authenticated Content Verification**", US Patent 18/779,858, Issued Dec. 2024 (Filed Jul. 2024)
10. Jiaxin Guan, Daniel Wichs, and Mark Zhandry, "**Cryptographic Data Message Expansion for Increasing Adversarial Storage Requirements**", US Patent Application 18/708,201, Allowed (National Phase, Filed May 2024)
11. Jiaxin Guan, Daniel Wichs, and Mark Zhandry, "**System and Method for Security Against Bounded-Storage Mass Surveillance**", US Patent Application 18/437,215, Published (Filed Feb. 2024)
12. Jiaxin Guan, Daniel Wichs, and Mark Zhandry, "**Cryptographic Data Message Expansion for Increasing Adversarial Storage Requirements**", PCT Patent Application PCT/US22/49033, Published (Filed Nov. 2022)